

1. A round-robin tennis tournament has taken place among n people, i.e. every pair of persons has played exactly one tennis game against each other. The results of this table are accessible to you in the form of a matrix M , where $M_{i,j} = 1$ if player i won against player j , and $M_{i,j} = 0$ otherwise. The entries $M_{i,i}$ are not relevant. An example is shown below.

	1	2	3	4
1		0	1	1
2	1		0	1
3	0	1		0
4	0	0	1	

An ordering of the players $i_1, i_2, \dots, i_n$ is called Hamiltonian if player i_j won against player i_{j+1} for every $j < n$. In the above example, 2, 1, 4, 3 is a Hamiltonian ordering. There can be more than one such ordering.

Describe an algorithm to find such an ordering by accessing the table at most $O(n)$ times. **[5 marks]**

2. Give asymptotic upper bounds for the functions described by the recurrences below.

(i) $T(n) = T(n/4) + T(3n/4) + O(n)$. **[3 marks]**

(iii) $f(n) = 3f(n/4) + O(n^{2/3})$. **[3 marks]**

3. (a) If $f(x)$ is a polynomial of degree 40 and $g(x)$ is a polynomial of degree 20, which primitive root of unity (i.e. ω_n for which n) would be suitable to use in the DFT-computation to find the product of $f(x)$ and $g(x)$? **[3 marks]**

(b) Find DFT_4 of the sequence (1, 1, 2, 3). **[3 marks]**

4. Describe an efficient algorithm that checks whether a given array $A[1], \dots, A[n]$ of n integers contains a Pythagorean triple, i.e. a, b, c such that $a^2 + b^2 = c^2$. **[6 marks]**

5. You are given a collection of intervals $[a_1, b_1], [a_2, b_2], \dots, [a_n, b_n]$, with $a_i < b_i$ being integers. Your goal is to find the number of pairs of intervals which intersect with each other. For example, if the input is $[1, 4], [-5, 3], [2, 3], [4, 5]$, then the output is 4. Describe a $O(n \log n)$ algorithm to solve this problem. **[6 marks]**

6. (a) Describe an algorithm that accepts an array $A[1..n]$ of distinct integers and an integer $k \in \{1, 2, \dots, n\}$, and finds the k smallest elements of A in $O(n)$ time. For example, if the array is $[4, -1, 23, 10, 15]$ and $k = 3$, the output consists of $4, -1, 10$ in some order. [3 marks]
- (b) Explain how to use (a) to solve the following problem: Given an array $A[1..n]$ of integers, an integer $k \in \{1, 2, \dots, n\}$ and another integer x , find the k elements of A that are closest to x (with respect to absolute difference with x). For example, if the array is $[3, 5, 1, 10, 7, 2, 7, 4, 4]$ and $x = 6$ and $k = 2$, then both $\{5, 7\}$ and $\{7, 7\}$ are correct. [3 marks]
7. You are in-charge of sorting the schedule of flights in a large airport. Your goal is to sort the flights in the following order of priority: departure time (hh:mm format), destination code (eg; HYD), flight number (an alphanumeric 6-letter code). Explain how to do this efficiently and correctly.
- [5 marks]